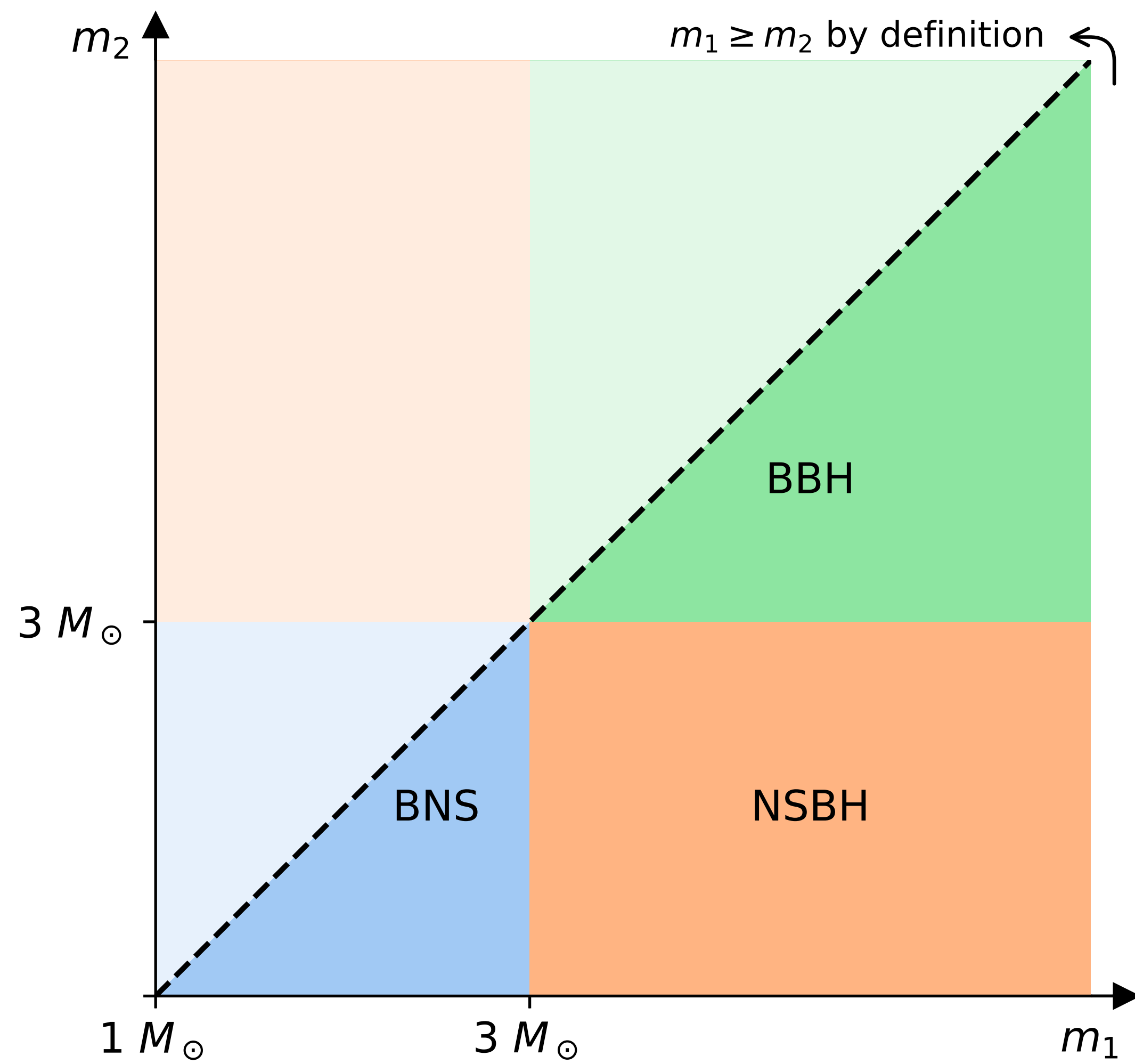


Digression on p_astro and EM_bright properties



Some useful definitions

Sky localization map: a file that contains a pixelated reconstruction of the joint distribution $P(\text{RA}, \text{Dec})$, plus a conditional posterior $P(D_L | \text{RA}, \text{Dec})$ on the luminosity distance. With these 3 info, we can fully build the 3D localization of the source

Sub Solar Mass search: there are now 2 pipelines (GstLAL and MBTA) that perform an additional search on mass ranges where we expect exotic objects:

$$0.2M_{\odot} < m_2 < 1.0M_{\odot} \text{ and } 0.2M_{\odot} < m_1 < 10M_{\odot}$$

Caveat: an SSM search can also trigger on a BNS!!

P(HasSSM) is also now released as part of IGWN alerts, if an SSM search triggers, reporting the probability that at least one has mass $< 1 M_{\text{sun}}$